

Digital Banking App - Data Model & Entity Relationship Design

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Version 1.0

1. Core Entities

Entity: User - Fields: user_id (UUID PK), email (VARCHAR 255 UNIQUE, encrypted), phone (VARCHAR 20, encrypted), first_name (VARCHAR 100, encrypted), last_name (VARCHAR 100, encrypted), dob (DATE, encrypted), ssn_hash (CHAR 64, SHA-256), address_id (UUID FK), kyc_status (ENUM: pending, verified, failed, manual_review), created_at (TIMESTAMP).

Entity: Account - Fields: account_id (UUID PK), user_id (UUID FK), account_type (ENUM: checking, savings, investment), account_number (VARCHAR 20, encrypted), routing_number (CHAR 9), balance (DECIMAL 15,2), available_balance (DECIMAL 15,2), currency (CHAR 3 DEFAULT 'USD'), status (ENUM: active, frozen, closed), opened_at (TIMESTAMP).

Entity: Transaction - Fields: transaction_id (UUID PK), account_id (UUID FK), type (ENUM: debit, credit, transfer, fee, interest, investment), amount (DECIMAL 15,2), balance_after (DECIMAL 15,2), merchant_name (VARCHAR 255 NULL), merchant_category (VARCHAR 50 NULL), description (VARCHAR 255), status (ENUM: pending, posted, failed, reversed), created_at (TIMESTAMP).

Entity: Card - Fields: card_id (UUID PK), account_id (UUID FK), card_number_encrypted (BYTEA), expiry_month (SMALLINT), expiry_year (SMALLINT), cvv_encrypted (BYTEA), card_type (ENUM: virtual, physical), status (ENUM: active, frozen, cancelled, expired), daily_limit (DECIMAL 10,2 DEFAULT 5000), issued_at (TIMESTAMP).

Entity: Transfer - Fields: transfer_id (UUID PK), sender_account_id (UUID FK), recipient_id (UUID FK NULL), recipient_external (JSONB NULL), amount (DECIMAL 15,2), fee (DECIMAL 8,2 DEFAULT 0), memo (VARCHAR 100 NULL), method (ENUM: internal, fednow, ach), status (ENUM: pending, processing, completed, failed), settled_at (TIMESTAMP NULL).

Entity: Investment - Fields: investment_id (UUID PK), user_id (UUID FK), ticker (VARCHAR 10), shares (DECIMAL 15,8), avg_cost_basis (DECIMAL 15,4), current_price (DECIMAL 15,4), unrealized_pl (DECIMAL 15,2), last_updated (TIMESTAMP).

Entity: SavingsGoal - Fields: goal_id (UUID PK), account_id (UUID FK), name (VARCHAR 100), target_amount (DECIMAL 15,2), current_amount (DECIMAL 15,2), target_date (DATE NULL), auto_deposit_amount (DECIMAL 10,2 NULL), auto_deposit_frequency (ENUM: daily, weekly, biweekly, monthly NULL), created_at (TIMESTAMP).

2. Relationships

User 1:N Account - A user can have checking, savings, and investment accounts.

Account 1:N Transaction - Each account has a full transaction ledger.

Account 1:N Card - An account can have multiple cards (virtual + physical).

User 1:N Transfer - A user sends many transfers over time.

User 1:N Investment - A user holds multiple stock/ETF positions.

Account 1:N SavingsGoal - A savings account can have multiple goals.

3. Data Security & Compliance

REQ-FIN-083: All PII fields encrypted at rest with AES-256 via AWS KMS. Encryption keys rotated quarterly.

REQ-FIN-084: SSN stored only as SHA-256 hash. Full SSN transmitted to KYC provider but never stored.

REQ-FIN-085: Card numbers (PAN) tokenized using Visa Token Service. Raw PAN never stored in application database. PCI DSS Level 1 compliance required.

REQ-FIN-086: All financial transactions logged to immutable audit ledger with user_id, action, amount, timestamp, source_ip. Retained for 7 years per BSA/AML regulations.

REQ-FIN-087: SAR (Suspicious Activity Report) auto-generation for transactions matching OFAC/AML patterns. Alert threshold: \$10,000 single transaction or \$5,000 cumulative in 24 hours.

4. Data Partitioning & Archival

REQ-FIN-088: Transaction table partitioned by month using PostgreSQL range partitioning. Active partition: current + 2 months. Older partitions moved to warm storage.

REQ-FIN-089: Real-time balance maintained as materialized view refreshed on each transaction. Reconciliation job runs daily at 03:00 UTC.

REQ-FIN-090: Investment positions updated real-time during market hours via WebSocket price feeds. Batch valuation at market close (4pm ET).